SatTrack

By: Nathania Castillo
Colby Campbell
Ryan Hashemian
Ashton Giles

Source: EcoOnline, “Benefits of Satellite Tracking”

Project Motivation

Source: Fun Kids – Amy's Aviation

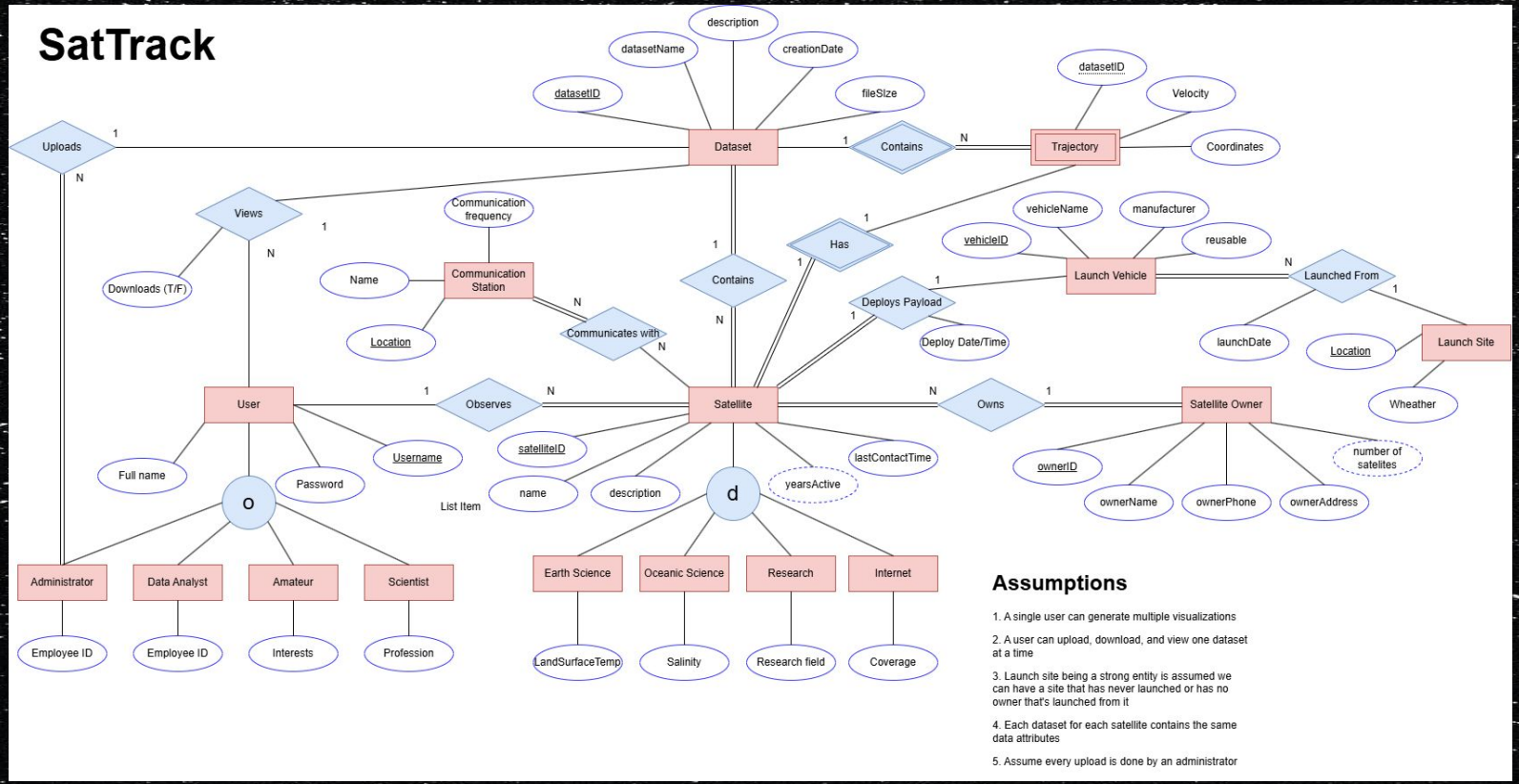
Motivation:

Our project aims to increase public awareness of space exploration and highlight humanity's scientific progress. Our goal is to inspire interest for users who are not well informed about space; and perhaps inspire research within our educated user base. The final project will be an interactive web application to view and interact with satellite data.

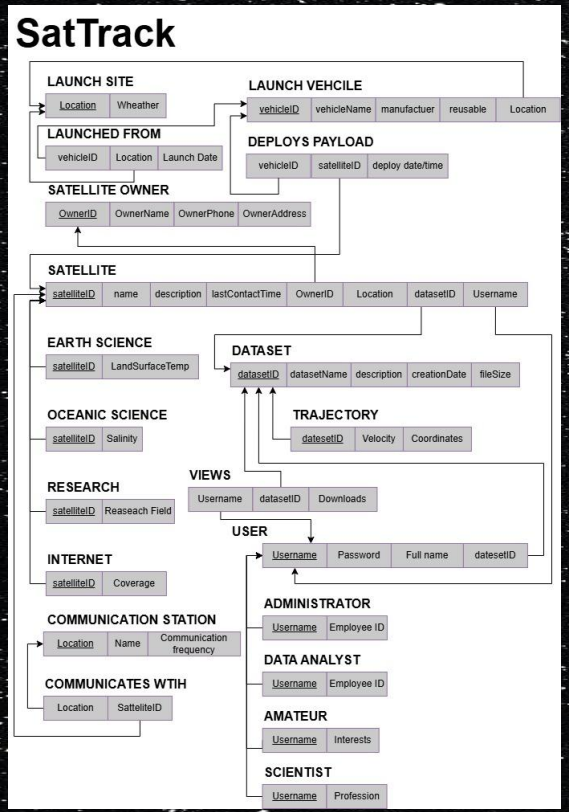
Project Type: Web-Based Application

Source: Rudzhan (Adobe Stock)

EERD


SatTrack
description
datasetName
creationDate
datasetID
fileSize
Dataset
Contains
Trajectory
datasetID
Velocity
Coordinates
Uploads
Views
Downloads (T/F)
Communication frequency
Name
Communication Station
Location
Communicates with
Contains
Has
vehicleName
manufacturer
vehicleID
reusable
Launch Vehicle
Launched From
Deploys Payload
Deploy Date/Time
launchDate
Location
Launch Site
Wheather
User
Observes
Satellite
Owns
Satellite Owner
Full name
Password
Username
satelliteID
name
description
List Item
lastContactTime
yearsActive
ownerID
ownerName
ownerPhone
ownerAddress
number of satelites
o
d
Administrator
Data Analyst
Amateur
Scientist
Employee ID
Employee ID
Interests
Profession
Earth Science
Oceanic Science
Research
Internet
LandSurfaceTemp
Salinity
Research field
Coverage
Assumptions
1. A single user can generate multiple visualizations
2. A user can upload, download, and view one dataset at a time
3. Launch site being a strong entity is assumed we can have a site that has never launched or has no owner that's launched from it
4. Each dataset for each satellite contains the same data attributes
5. Assume every upload is done by an administrator

Relational Model



Assumptions

1. For the satellite superclass and subclass relationship we chose to create a relation for each subclass and pass the primary key of the superclass to the subclasses

2. An assumed constraint for the User superclass subclass relationship is that while a user can have overlapping user types we assume an administrator is independent of other roles since its access level is highest

3. New relations added LAUNCH VEHICLE, DEPLOYS PAYLOAD, COMMUNICATION STATION, COMMUNICATES WITH, implemented in the EER as well building off feedback from the EER diagram

4. Administrator has a relationship with dataset as administrators will be the only ones allowed to upload datasets

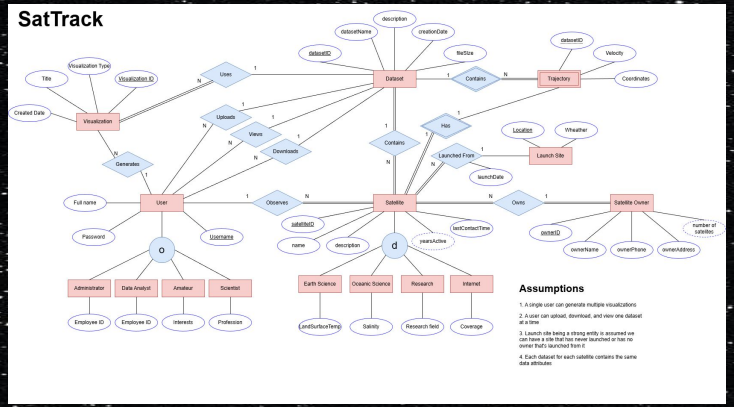
Project Feedback Discussion

Feedback from Initial EER Diagram:

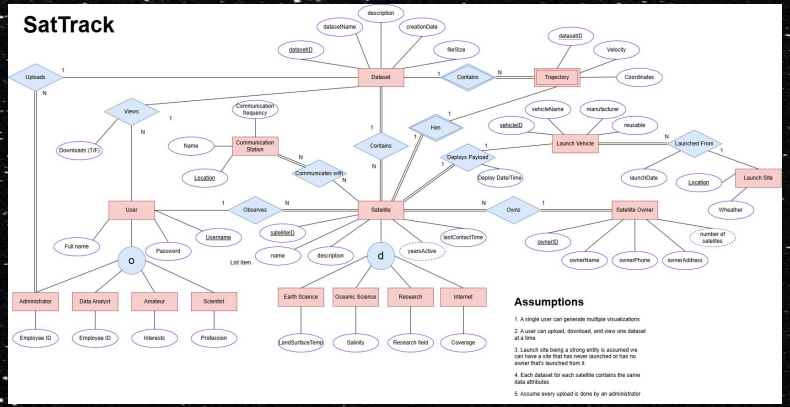
- Remove the Visualization entity
- Fixed the user types to effectively reflect separation of privileges between administrative users and regular users
- Added new entities for satellite communication stations and launch vehicles for satellites to EER and RM diagrams

After Updating the EER diagram with the feedback we applied this feedback during the conversion between EER and RM for progress report two.

Before



After



Project Plan and Tech Stack

Tech Stack:

- Frontend: React (JavaScript)
- Backend: Django (Python)
- Database Management System: MySQL

Development Tools:

- Shortcut (Task Management)
- Figma (Frontend Design)

References

EcoOnline. (2024). *Benefits of satellite tracking*. EcoOnline. https://www.ecoonline.com/en-ca/blog/benefits-of-satellite-tracking/

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Rudzhan. (n.d.). *Set of five diverse thoughtful people with puzzled or quizzical expressions and floating question marks on white* [Vector illustration]. Adobe Stock. https://stock.adobe.com/ca/images/set-of-five-diverse-thoughtful-people-with-puzzled-or-quizzical-expressions-and-floating-question-marks-on-white-colored-vector-illustration/356519275

Thank You

Source: ESA Education (2022)